

Virtual Refrigeration System Report

Generated: 2025-12-05 10:12:27

System Information

Refrigerant:	R134a
Model Type:	single-stage
Evaporator Temp:	-10 °C
Condenser Temp:	40 °C
Superheating:	5 °C
Subcooling:	5 °C
Compressor Eff:	85 %

State Points

Point	P (kPa)	T (°C)	h (kJ/kg)	s (kJ/kg·K)	Quality
1	235.40	-5.00	395.00	1.7815	N/A
2	1137.70	66.97	466.97	2.0193	N/A
3	1137.70	35.00	249.00	1.1688	N/A
4	235.40	-10.00	249.00	1.1995	0.309

Performance Parameters

Parameter	Value	Unit
Cop	2.029	-
Compressor Power	4.929	kW
Compressor Work	71.97	kW
Cooling Capacity	10	kW
Heat Rejected	217.97	kW
Mass Flow Rate	0.06849	kg/s
Refrigeration Effect	146	kW
Volumetric Efficiency	80.8	%